MD FAHIM HASAN

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Education

Colorado State University
Ph.D., Civil and Environmental Engineering

Missouri University of Science and Technology
Masters, Geological Engineering

Bangladesh University of Engineering and Technology
Bachelors, Civil Engineering

Technical Skills

- **Programming** : Python, R
- **Machine/Deep Learning** : scikit-learn, PyTorch
- **Geospatial** : XArray, Dask, Rasterio, GDAL, GeoPandas
- **Remote Sensing** : STAC, Google Earth Engine
- **Cloud & HPC** : AWS SageMaker, Google BigQuery, Linux, HPC
- **IDEs** : PyCharm, JupyterLab, Posit Workbench, Spyder
- **Version Control** : GitHub, GitLab
- **GIS** : ArcGIS Pro, QGIS
- **Modeling** : Energy balance ET modeling (TSEB), SWAT, HEC-RAS

Experience

Agricultural Data Scientist
Syngenta Group | September 2026 – Present

Graduate Research Assistant
Remote Sensing Hydrology Group, Colorado State University | Aug 2022 – Aug 2026
NASA Project

- Developed a physics-constrained machine learning (ML) model to estimate effective precipitation over irrigated agricultural areas in the Western U.S.
- Built end-to-end pipelines combining satellite-based inputs and in-situ data for groundwater pumping prediction using process-based and deep learning (DL) models.
- Queried and processed large-scale datasets in Google Earth Engine (GEE), covering 17 states of the Western U.S., and integrated with other geospatial data sources.
- Designed and deployed automated geospatial python workflows on High Performance Computing (HPC) environments for ML/DL.
- Maintained version-controlled open-source codebases and databases to ensure reproducibility and scalability for geospatial ML/DL models in Git.

Agricultural Systems Data Scientist Co-Op
Syngenta Group | May 2025 – Aug 2025

- Developed and validated a Two-Source Energy Balance (TSEB) model using UAV thermal, NIR, and RGB imagery to map evapotranspiration (ET).
- Designed and implemented field deployments of environmental sensors including Sentek Drill & Drop, Irrometer Watermark, Phytech, Davis weather stations, and LICOR-600 Porometer/Fluorometer.
- Integrated and analyzed multi-sensor datasets alongside modeled ET to evaluate water use efficiency (WUE) across soil treatments.

- Built an end-to-end Python pipeline for UAV image processing, TSEB modeling, and water use efficiency (WUE) evaluation for different soil treatments.
- Emphasized reproducibility, modularity, and scalability via Git-based version control. The developed pipeline enabled TSEB model deployment from small plots to the farm scale.

Vegetable R&D Data Science Intern

Bayer Crop Science | May 2023 – Aug 2023

- Developed a machine and deep learning framework to downscale weather datasets (e.g., temperature, precipitation) for environmental modeling.
- Managed multi-source datasets (e.g., weather, soil, elevation) and built a harmonized analysis-ready dataset (ARD) in AWS.
- Conducted cloud-based data querying and processing using tools like Pandas-gbq, BigQuery, and h3-indexed spatial analysis.
- Ensured data reproducibility through automated python workflows and Git-based pipeline versioning.

Graduate Research Assistant

Missouri University of Science and Technology | Jan 2021 – Jul 2022

NGA Project

- Built a machine learning-based (classification) global land subsidence model by fusing geospatial and remote sensing data to map areas susceptible to groundwater extraction.
- Extracted training features from InSAR and other satellite products; integrated in model workflow; trained and validated models across diverse geographies.

USGS Project

- Collaborated on the Aquaculture and Irrigation Water-Use Model (AIWUM)-2.0 machine learning model to estimate groundwater use for crop irrigation in the Mississippi Alluvial Plain.
- Developed spatial linkages between pumping wells and crop fields using scikit-learn's KDTree algorithm.
- Co-developed improved input features for the updated AIWUM-2.0 model, enhancing prediction accuracy.

Bangladesh Water Development Board

Assistant Engineer | Oct 2018 – Dec 2020

- Supervised field operations for various water resource engineering projects, ensuring compliance with technical standards and timelines. Assisted in financial planning and budget preparation for infrastructure and water management initiatives.
- Conducted data analysis to support project planning, performance evaluation, and resource optimization.
- Contributed to the implementation and policy review of the Ganges Water Sharing Treaty.

Publications/Abstracts

Peer Reviewed –

[1] **Hasan, M. F**., Smith, R. G., Majumdar, S., Huntington, J. L., Alves Meira Neto, A., & Minor, B. A. (2025). Satellite data and physics-constrained machine learning for estimating effective precipitation in the Western United States and application for monitoring groundwater irrigation. Agricultural Water Management, 319, 109821.

[2] Majumdar, S., Smith, R., **Hasan, M.F**., Wilson, J. L., White, V.E., Bristow, E.L., Rigby, J.R., Kress, W., and Painter, J.A. (2024). Improving Crop-Specific Groundwater Use Estimation in the Mississippi Alluvial Plain: Implications for Integrated Remote Sensing and Machine Learning Approaches in Data-Scarce Regions. Journal of Hydrology: Regional Studies, 52, 101674.

[3] **Hasan, M. F.**, Smith, R., Vajedian, S., Pommerenke, R., & Majumdar, S. (2023). Global land subsidence mapping reveals widespread loss of aquifer storage capacity. Nature Communications, 14(1), 6180.